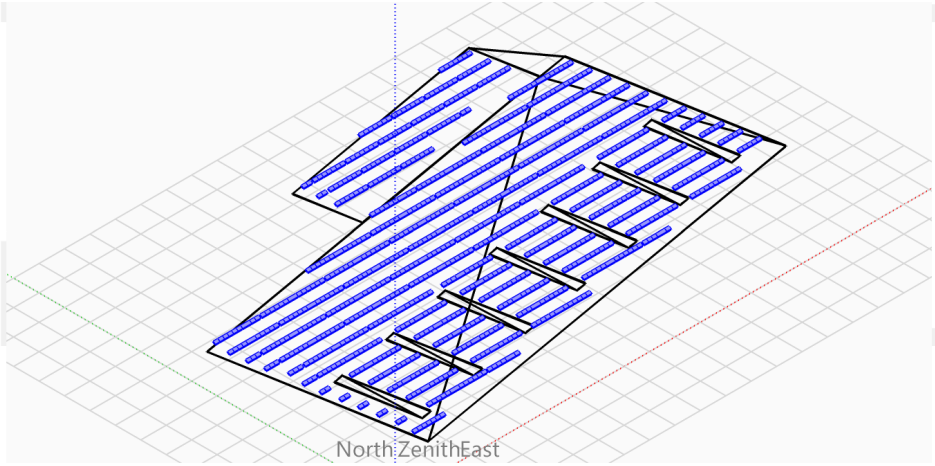
512 kWp rooftop PV for a logistics centre

From an empty roof to a simulated, wired system: site analysis, row geometry, string and inverter sizing, cable design, a 3D near-shading scene and a full PVsyst yield simulation.

Site — GEIS Logistics Center, Nürnberg region

Tools — Hand calculation · AutoCAD · Blender · PVsyst 8.1



3D near-shading scene — roof, skylights and 1,078 modules (PVsyst)

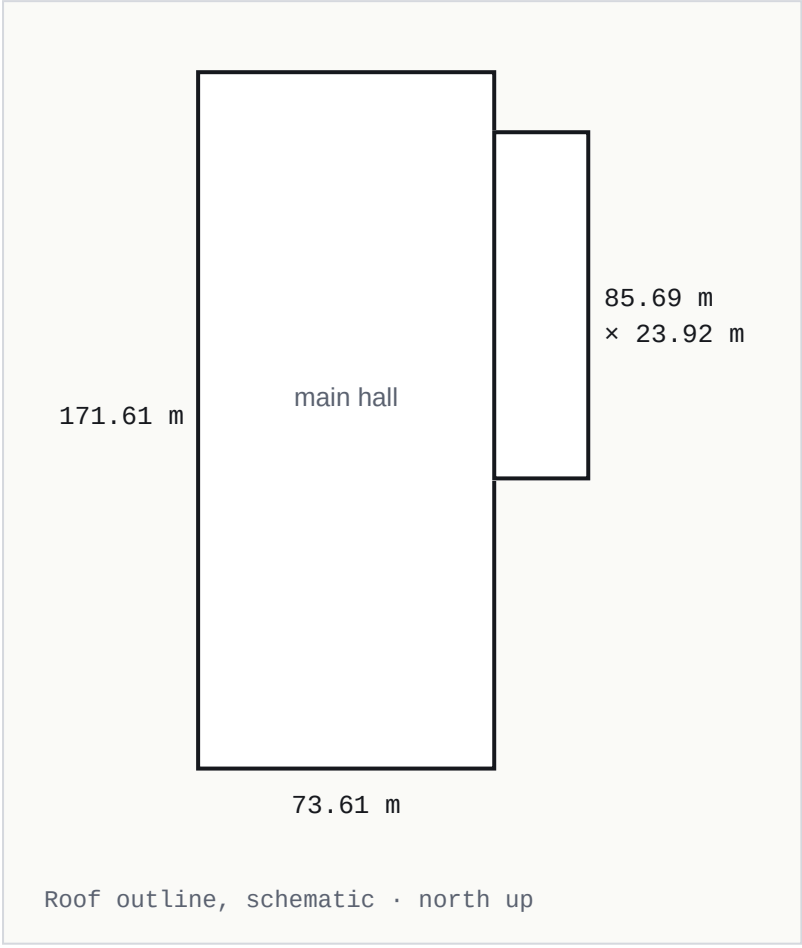
512 kWp	440 kW	1,078	584 MWh	1,141	84.2 %
DC capacity	AC · 4 × SMA STP110	modules · 475 Wp	per year (year 1)	kWh/kWp specific yield	performance ratio

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Self-initiated concept study on a real building; not commissioned by the owner. Not for construction.

A 1.47 ha flat roof, minus skylights and edge zones



Roof outline, schematic · north up

Gross roof area	14,681.91 m²
Skylights, 7 × (29 × 3 m)	− 609.00 m²
Available area	**14,072.91 m²**

SITE & CLIMATE

Location	Nürnberg region
Altitude	≈ 310 m
Weather data	Meteonorm 9.0
Global horizontal	1,158 kWh/m²
Design T_{min} / T_{max}	−21.4 / 35.8 °C
Albedo	0.20

ROOF CONSTRAINTS

Edge setback	12 m
Skylight rows	29 × 3 m
Skylight spacing	25 m
Mounting	tilted tables
Tilt / azimuth	36° / 0° (south)
Thermal U_c	29 W/m²K

PV MODULE — TRINA VERTEX S+ TSM-NEG9RH.28, 475 WP

Size 1.762 × 1.134 m	Power density 237.7 Wp/m²	V_{oc} / V_{mpp} 55.0 / 46.4 V	I_{sc} / I_{mpp} 10.93 / 10.24 A
Temp. coeff. β −0.24 %/K	NOCT 43 °C	Area per module 1.998 m²	Mounted as landscape, 1 row

Design temperatures come from the site climate data: the cold extreme sets the maximum string voltage, the hot extreme sets the minimum MPPT voltage. The open-rack thermal factor (29 W/m²K) reflects ventilated tilted tables rather than a roof-integrated array.

Estimate by hand, model in CAD, verify by simulation

Every key number was first derived on paper, then checked against a detailed layout and an hourly PVsyst simulation. Where the two disagreed, the reason is documented (sheet 12).

Estimate

hand calculation

01 Site survey
Roof and skylight areas, setbacks, design temperatures

02 Row geometry
Portrait vs landscape; pitch from a 7° winter sun angle

03 Module count
First estimate of what fits: 1,028 modules, 488 kWp

04 String sizing
Voltage window at −21 °C and +65 °C cell → 14 modules

05 Inverters & MPPT
4 × SMA STP110-60, string allocation, DC/AC ratio

Model

CAD + 3D

06 Detailed layout
Real rows drawn at 6.30 m pitch around the skylights: 1,078 modules, 77 strings

07 Near-shading scene
AutoCAD solids → STL → Blender → DAE → PVsyst 3D scene

Why the layout fits more (+5 %)

The area formula divides by the full module area (2.0 m²), although a tilted row only takes its projected depth — so it keeps about 19 % of the roof in reserve. The drawn rows use the real row lengths and lose only about 15 % to skylights, edges and row ends. Strings and inverters were re-allocated to the new count.

Verify

simulation + electrical

08 Yield simulation
PVsyst hourly model: 584 MWh/yr, PR 84.2 %, full loss chain

09 Cable sizing
DC 6 mm² and AC 4G150 mm²: ampacity and voltage drop

10 Single-line diagram
Generators, inverters, protection and grid connection point

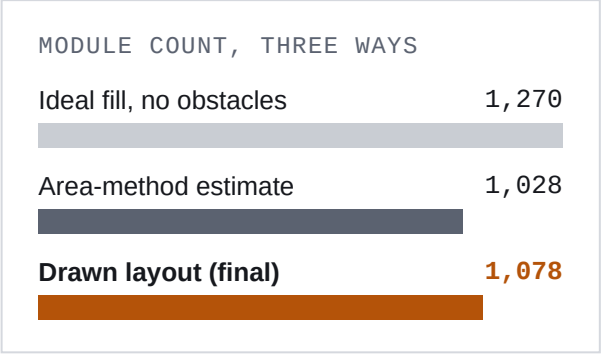
11 Cross-check
Hand calculation vs simulation, deviations explained

12 Economics
Excel model: LCOE, NPV, IRR, payback and sensitivities

Landscape rows cut the pitch from 9.86 m to 6.29 m

sun ray, 7° (winter design angle) 36° h 0.66 w 0.917 shadow 5.37 m pitch 6.29 m next row

$w = L \cdot \cos\beta \quad h = L \cdot \sin\beta \quad \text{shadow} = h / \tan\alpha_{sun} \quad \text{pitch} = w + \text{shadow}$



Portrait vs landscape (one module per table row)

Parameter (m)	Portrait	Landscape
Collector width L	1.762	1.134
Table width w	1.430	0.917
Table height h	1.035	0.660
Shadow at 7°	8.429	5.370
Pitch	9.859	**6.287**

Low landscape tables shade a much shorter strip behind them, so more rows fit on the same roof and wind exposure stays low.

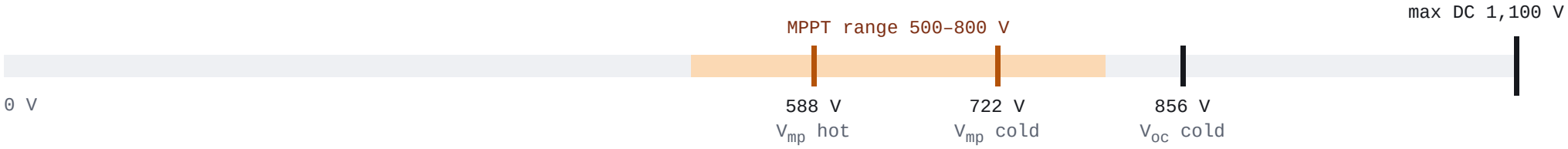
First estimate: how many modules fit?

Effective PV area = 14,072.91 × 0.917 / 6.287	2,052.6 m²
÷ module area 1.998 m²	1,028 modules
× 475 Wp	488.3 kWp

Why the drawn layout reached 1,078

The formula divides by the full module area (1.762 × 1.134 m), but on the roof a tilted module only takes 1.762 × 0.917 m. That builds in a ~19 % reserve. With no obstacles, 14,072.91 / 6.287 = 2,238 m of rows would hold 1,270 modules. The real layout loses ~15 % of that to skylights, edges and row ends — and ends at 1,078.

14 modules per string fit from −21 °C to +65 °C



String of 14 × TSM-NEG9RH.28 against the SMA STP110-60 input limits

Cold morning: upper limit

$T_{cell} = T_{amb,min} = -21.4\ °C$

$V_{oc,cold} = 55.0 \times [1 - 0.0024 \times (-46.4)]$

$= 61.13\ V$

$N_{max} = 1{,}100 / 61.13 = 17.99 \rightarrow \mathbf{17}$

$V_{mp,cold} = 46.4 \times 1.111 = 51.57\ V$

Hot afternoon: lower limit

$T_{cell} = 35.77 + (43 - 20)/0.8 = 64.5\ °C$

$V_{mp,hot} = 46.4 \times [1 - 0.0024 \times 39.5] = 42.0\ V$

$N_{min} = 500 / 42.0 = 11.9 \rightarrow \mathbf{12}$

Checks for N = 14

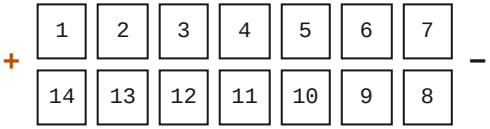
Condition	String	Limit
$V_{oc,cold} \times 14$	855.8 V	< 1,100 V
$V_{mp,hot} \times 14$	588 V	> 500 V
$V_{mp,cold} \times 14$	722 V	< 800 V
I_{sc} per string input	10.93 A	< 20 A

String power, 14 × 475 Wp 6.65 kWp

Current limit basis 40 A / MPPT, 2 strings

14 sits in the middle of the 12–17 range, leaving margin on both sides. The V_{oc} temperature coefficient was also applied to V_{mpp}, as no separate value was available.

ONE STRING

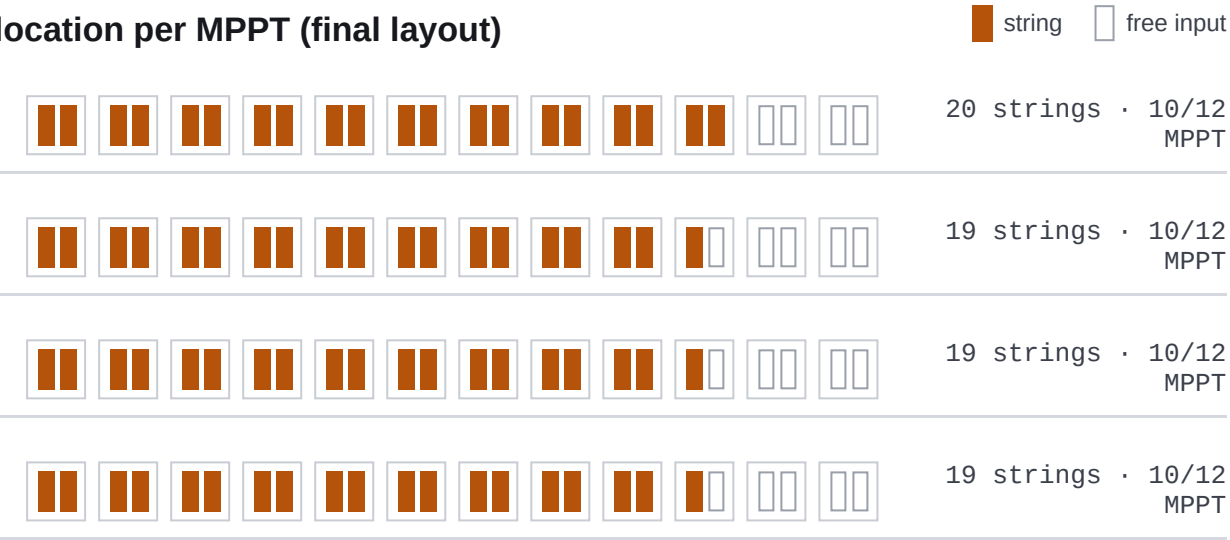


14 modules in series, folded over two table rows

Power	6.65 kWp
V_{mpp} at STC	649.6 V
V_{mpp} range	588–722 V
V_{oc} at −21 °C	855.8 V
I_{mpp} / I_{sc}	10.24 / 10.93 A
Strings in plant	77

Four 110 kW string inverters, 77 strings, DC/AC 1.16

String allocation per MPPT (final layout)



Inverter	Strings	Modules	DC kWp	DC/AC
INV01	20 × 14	280	133.0	1.21
INV02–04 (each)	19 × 14	266	126.4	1.15
Total	**77**	**1,078**	**512.1**	**1.16**

SMA SUNNY TRIPOWER CORE2 STP110-60

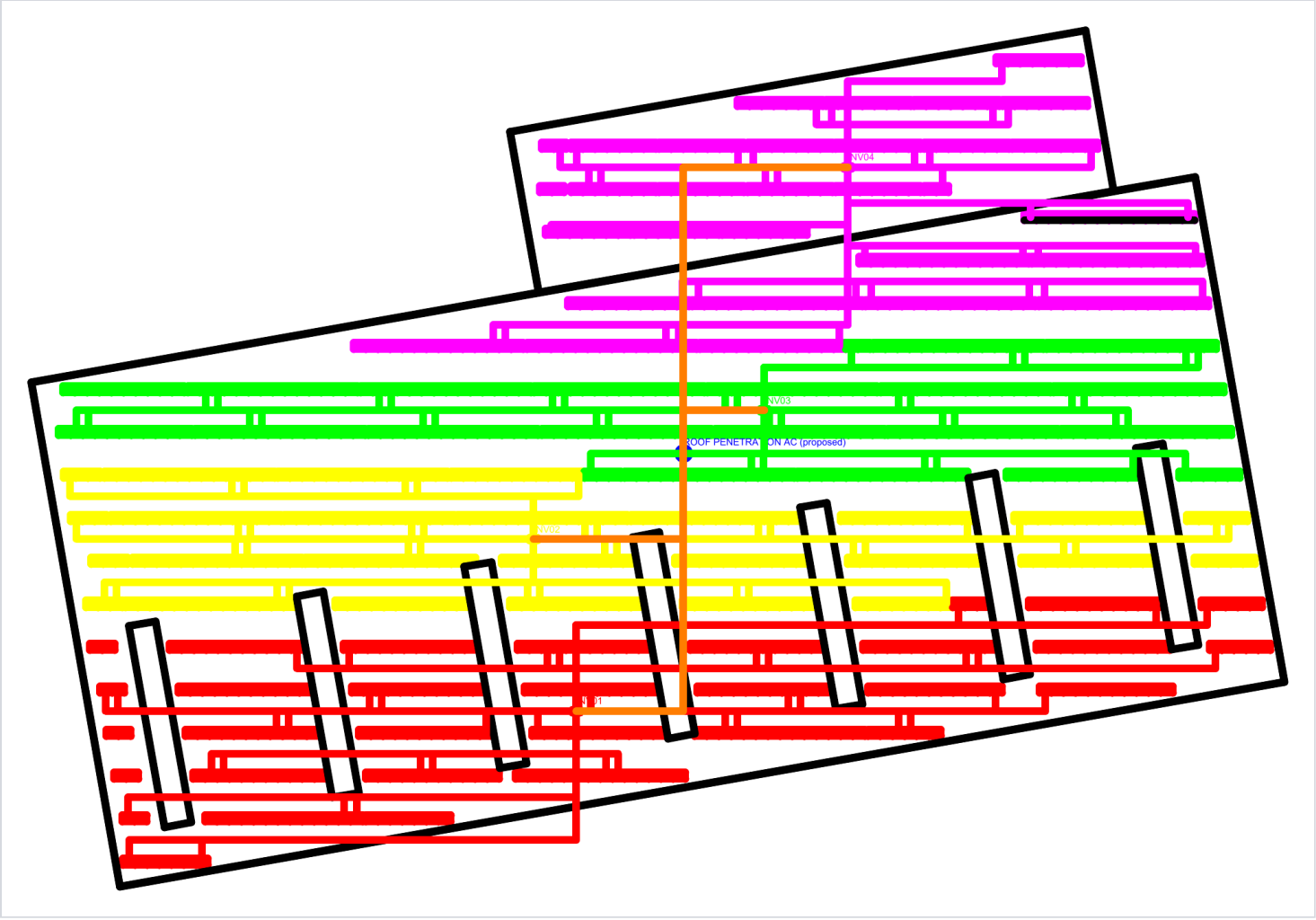
AC power	110 kW, 400 V
AC current	158.8 A
MPP trackers	12 × 2 inputs
MPPT range	500–800 V
Max DC voltage	1,100 V
Max DC per unit	165 kWp

ESTIMATE → FINAL LAYOUT

	estimate	final
Strings	73	77
Modules	1,022	1,078
DC power	485.5	512.1 kWp
DC/AC ratio	1.10	1.16

Every inverter stays well under its 165 kWp DC limit and keeps 2 MPPTs free. The higher DC/AC ratio raises annual yield but brings 3.7 % clipping loss in the simulation — a trade-off flagged for optimisation on sheet 15.

Each inverter gets its own block of rows



AutoCAD plan GEIS_PV_v3 — module rows by inverter, skylights, AC routes (schematic)

LEGEND

INV01	280 mod.
INV02	266 mod.
INV03	266 mod.
INV04	266 mod.
AC feeders	4G150
Skylights	7 × 29 m

Contiguous fields. Every inverter feeds one block of neighbouring rows, so string cables stay short — the longest is 111 m.

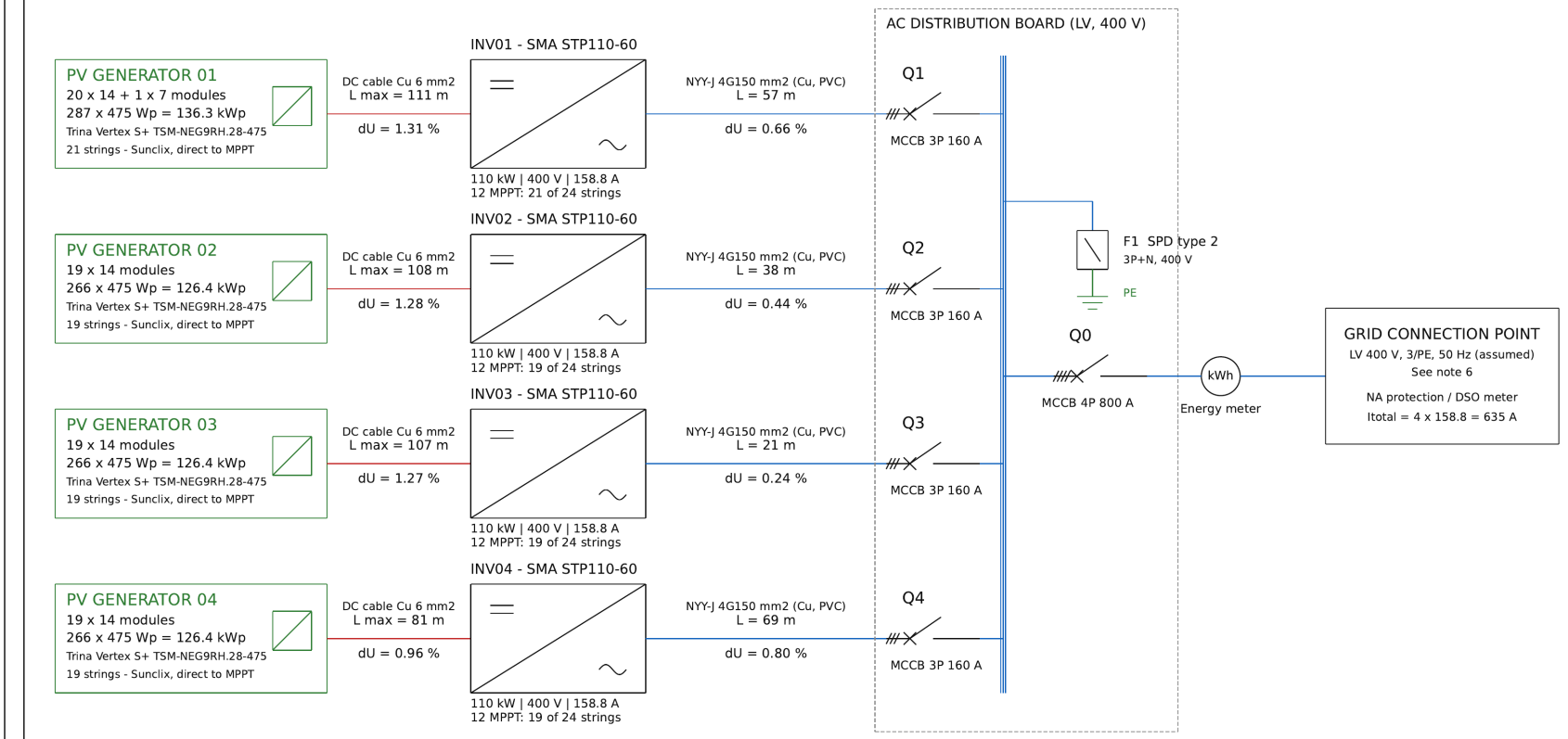
Skylights kept clear. Rows break around the seven skylight strips instead of shading them.

One roof penetration. All four AC feeders meet at a single proposed penetration, 21–69 m from the inverters.

77
strings of 14

111 m
longest DC run

4 generators → 4 inverters → LV board → grid



NOTES / DESIGN DATA

1. Modules: Trina Vertex S+ TSM-NEG9RH.28-475, 475 Wp, 14 in series (Vmpp 649.6 V, Impp 10.24 A, Isc 10.93 A).
2. Inverters: 4 x SMA Sunny Tripower CORE2 STP110-60 (110 kW, 400 V, 158.8 A, 12 MPPT); 1085 modules, 515.4 kWp installed.
3. DC cable Cu 6 mm2: Iz = 63 x 0.58 x 0.79 = 28.9 A >= 1.25 x Isc = 13.7 A; worst dU 1.31 % (INV01, L = 111 m).
4. AC cable NYY-J 4G150 Cu/PVC: Iz = 326 x 0.75 x 0.71 = 174 A; Ib 158.8 A < In 160 A < Iz 174 A.
5. AC voltage drop: INV01 0.66 %, INV02 0.44 %, INV03 0.24 %, INV04 0.80 % (limit 5 %).
6. Grid connection (LV main board, Q0 800 A) is an ASSUMPTION. A 440 kW plant needs DSO approval and VDE-AR-N 4110.
7. The 7-module string (3.3 kWp) sits on its own MPPT of INV01; PVsyst covers 1078 modules (512 kWp).
8. Cable lengths include +3 m per end; routes in the CAD plan are schematic.

PROJECT: GEIS Logistics Center, Nuernberg region

DRAWING: Single-line diagram, PV system 512 kWp

PVsyst (year 1): 584,175 kWh/a | 1141 kWh/kWp | PR 84.2 %

Sheet 1 / 1 Scale: not to scale Rev. A

Date: 2026-10-29

Designed by: Tayssir Almariri

TRIAL DESIGN - NOT FOR CONSTRUCTION

Ampacity first, then voltage drop: worst case 1.31 % DC, 0.80 % AC

DC string cables

Cu 6 mm², XLPE

$I_b = 1.25 \times I_{sc} = 1.25 \times 10.93 = 13.7$ A

$I_z = 63 \times 0.58_{temp} \times 0.79_{group} =$ **28.9 A** $\geq I_b$

$\Delta U = 2 \cdot \rho \cdot L \cdot I_{mpp} / S$, $U_{string} = 14 \times 46.4 = 649.6$ V

Longest string	L (m)	ΔU (V)	ΔU
INV01	111	8.52	**1.31 %**
INV02	108	8.29	1.28 %
INV03	107	8.22	1.27 %
INV04	81	6.22	0.96 %

Correction factors from IEC 60364-5-52 (B.52.14 temperature, B.52.20 grouping, method E), applied per IEC 62548. Strings connect directly to the MPPT inputs (no combiner box). $\rho_{Cu} = 0.0225$ Ω·mm²/m.

AC inverter feeders

NYY-J 4G150 mm²

$I_b = 158.8$ A (inverter rated current)

$I_z = 326 \times 0.71_{50\ °C} \times 0.75_{4\ cables} =$ **174 A**

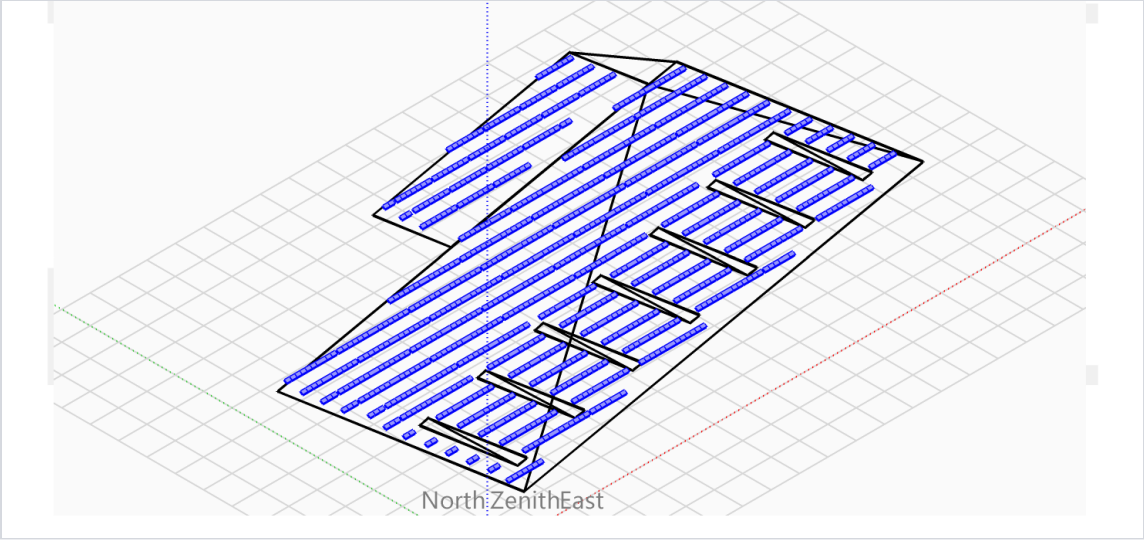
I_b 158.8 ≤ I_n 160 (MCCB) ≤ I_z 174 A

$u = (\rho \cdot L/S \cdot \cos\varphi + \lambda \cdot L \cdot \sin\varphi) \cdot I_b$, cos φ 0.8, λ 0.08 mΩ/m

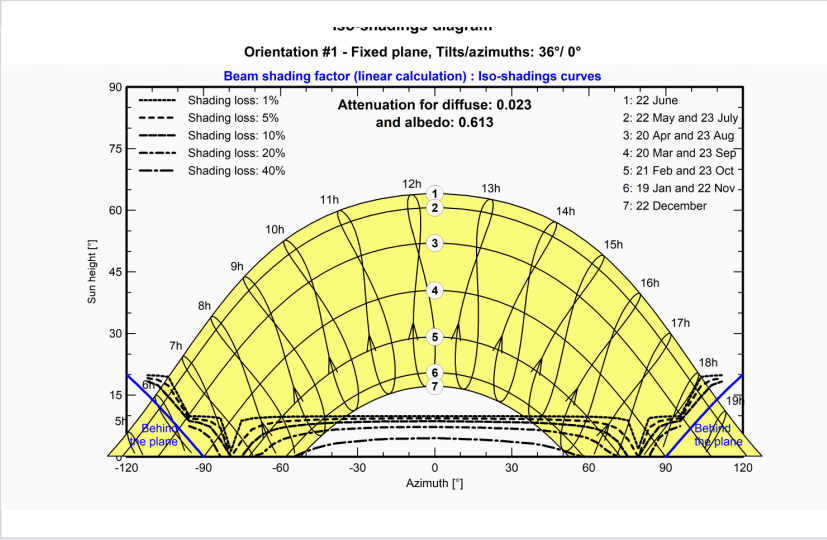
Feeder	L (m)	ΔU
INV01 → Q1	57	0.66 %
INV02 → Q2	38	0.44 %
INV03 → Q3	21	0.24 %
INV04 → Q4	69	**0.80 %**

Bars on both sides share one scale (100 px = 1 %). Cable lengths include +3 m per end; routes in the CAD plan are schematic. Board incomer Q0: MCCB 4P 800 A for 4 × 158.8 = 635 A.

From CAD solids to a 3D shading model: 1.6 % irradiance loss



PVsyst scene: roof, skylight strips and module rows



Iso-shading diagram — losses only at low winter sun

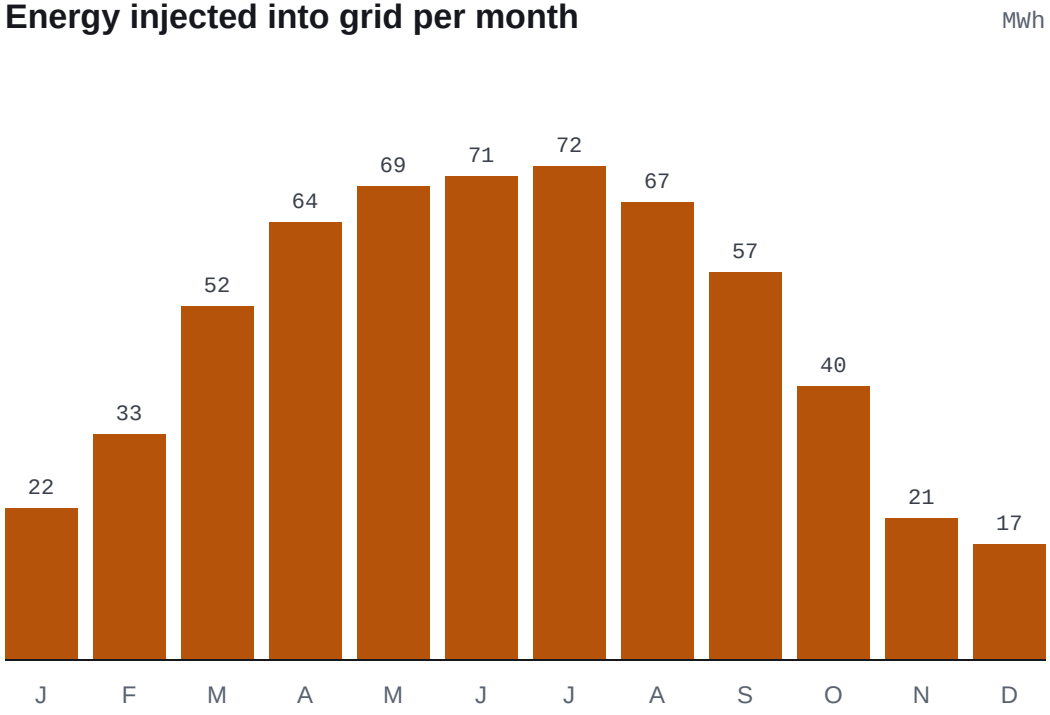
Model pipeline

AutoCAD 3D solids → STL (panels + structure) → Blender — PV material on top faces → DAE export → **PVsyst near shadings**

−1.61 %	7.1°	6.30 m	0.023
near-shading irradiance loss	shading limit angle (hand calc: 7°)	modelled row pitch	diffuse attenuation factor

584 MWh a year at a performance ratio of 84.2 %

Energy injected into grid per month



584,175
kWh/yr to grid

1,141
kWh/kWp/yr

1,356
kWh/m² on the 36° plane

PR stays 0.81–0.91 across the year; summer months dip from higher cell temperature and clipping.

Where the energy goes

IRRADIANCE		→ 1,283 kWh/m² effective
Near shadings	−1.61 %	
Soiling	−1.00 %	
Incidence angle (IAM)	−2.85 %	
PV ARRAY		→ 621.2 MWh at MPP
Irradiance level	−1.17 %	
Temperature	−1.27 %	
LID	−1.00 %	
Mismatch	−2.02 %	
DC wiring	−0.31 %	
Degradation, year 1	−0.20 %	
INVERTER & AC		→ 584.2 MWh to grid
Inverter efficiency	−1.99 %	
Clipping (over nominal power)	−3.68 %	
AC wiring	−0.35 %	

Largest single loss: inverter clipping (−3.68 %), a direct result of the 1.16 DC/AC ratio. Tilting to 36° gains +17 % irradiance over horizontal.

Paper estimates landed within 5 % of the detailed model

Quantity	Hand calculation	Layout / PVsyst	Δ	Why they differ
Row pitch	6.287 m	6.30 m	+0.2 %	Table height rounded to 0.66 m on paper
Shading limit angle	7.0°	7.1°	—	Follows directly from the modelled pitch
Specific yield	1,127 kWh/kWp	1,141 kWh/kWp	+1.2 %	Global Solar Atlas PVOUT (3.088 kWh/kWp/day) vs hourly PVsyst model with site losses
Modules	1,028	1,078	+4.9 %	Formula divides by full module area, not the tilted footprint (~19 % reserve); real rows lose only ~15 % to skylights and edges
DC power / DC-AC	488 kWp / 1.11	512 kWp / 1.16	+4.9 %	Same inverters, more modules
Ground coverage	14.6 %	18.0 %	def.	Different definitions: projected width / pitch (0.917 / 6.287) vs collector width / pitch (1.134 / 6.30)
DC wiring loss	1.31 % worst	0.51 % at STC	—	Worst single string vs array-wide average resistance
AC voltage drop	0.80 % (69 m)	0.67 % at STC	—	Hand calc at rated current, cos φ 0.8; PVsyst at STC operating point

Geometry holds

The 7° design-angle method reproduces the simulated pitch and limit angle almost exactly.

Yield is credible

An independent atlas estimate and the full hourly model agree within 1.2 %.

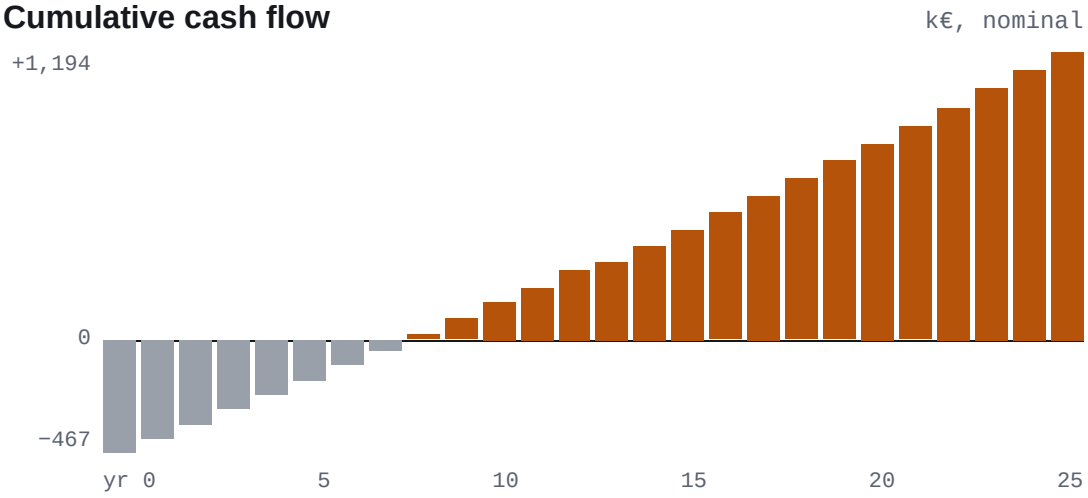
Layout beats the rule

Area formulas give a fast first number; a drawn layout is still needed for the real count.

Solar at 8.7 ct/kWh against 17.1 ct/kWh from the grid

467 k€	8.7 ct	7.7 yr	442 k€	12.7 %
investment (net)	LCOE per kWh	payback (9.8 discounted)	NPV, 25 yr at 5 %	IRR

Cumulative cash flow



NPV (k€) — the self-consumption lever

system cost €/kWp →

self-cons.	800	850	900	950	1,000
30 %	123	91	58	26	-6
40 %	251	219	186	154	122
50 %	379	347	314	282	250
60 %	507	475	**442**	410	378
70 %	635	603	570	538	506
80 %	763	731	698	666	634
90 %	891	859	826	794	762

Assumptions (all prices net, 2026)

System cost, turnkey	900 €/kWp	elektronik-zeit 2026
Grid protection > 100 kW	6,500 €	elektronik-zeit 2026
O&M + insurance	1.5 % + 10 €/kWp	LfL Bayern 02/2026
Inverter replacement, yr 13	70 €/kW	assumption
Self-consumption share	**60 %**	assumption, no load data
Avoided grid price	17.1 ct, +2 %/yr	BDEW 08/2026
Export (direct marketing)	5.0 ct	elektronik-zeit 2026
Negative-price loss on export	7.5 %	plant ≥ 400 kW
Yield / degradation	584 MWh, 0.4 %/yr	PVsyst
Discount rate / lifetime	5 % / 25 yr	KfW 270 + equity

Every kWh used on site saves 17.1 ct, every kWh exported earns ~4.6 ct. At 900 €/kWp the NPV stays positive down to about 25 % self-consumption — a measured load profile is the next step to pin this down.

Year-by-year cash flow and formulas: sheet 14.

Year by year: the investment is back in year 8

n	Year	Energy MWh	Self-used	Exported	Savings €	Export €	Costs €	Net €	Cumulative €
0	2026	–	–	–	–	–	–	**−467,345**	−467,345
1	2027	584.2	350.5	233.7	59,937	10,807	12,131	**58,613**	−408,732
2	2028	581.8	349.1	232.7	60,891	10,764	12,373	**59,282**	−349,450
3	2029	579.5	347.7	231.8	61,860	10,721	12,621	**59,960**	−289,490
4	2030	577.2	346.3	230.9	62,845	10,678	12,873	**60,650**	−228,840
5	2031	574.9	344.9	230.0	63,845	10,635	13,131	**61,350**	−167,490
6	2032	572.6	343.6	229.0	64,862	10,593	13,393	**62,062**	−105,428
7	2033	570.3	342.2	228.1	65,894	10,550	13,661	**62,784**	−42,644
8	2034	568.0	340.8	227.2	66,944	10,508	13,934	**63,517**	20,873
9	2035	565.7	339.4	226.3	68,009	10,466	14,213	**64,263**	85,136
10	2036	563.5	338.1	225.4	69,092	10,424	14,497	**65,019**	150,155
11	2037	561.2	336.7	224.5	70,192	10,383	14,787	**65,787**	215,942
12	2038	559.0	335.4	223.6	71,309	10,341	15,083	**66,568**	282,510
13	2039	556.7	334.0	222.7	72,445	10,300	46,185	**36,560**	319,070
14	2040	554.5	332.7	221.8	73,598	10,259	15,692	**68,164**	387,234
15	2041	552.3	331.4	220.9	74,770	10,218	16,006	**68,981**	456,215
16	2042	550.1	330.1	220.0	75,960	10,177	16,326	**69,810**	526,025
17	2043	547.9	328.7	219.2	77,169	10,136	16,653	**70,652**	596,678
18	2044	545.7	327.4	218.3	78,398	10,095	16,986	**71,507**	668,185
19	2045	543.5	326.1	217.4	79,646	10,055	17,326	**72,375**	740,560
20	2046	541.3	324.8	216.5	80,914	10,015	17,672	**73,257**	813,817
21	2047	539.2	323.5	215.7	82,202	9,975	18,026	**74,151**	887,968
22	2048	537.0	322.2	214.8	83,511	9,935	18,386	**75,059**	963,028
23	2049	534.9	320.9	213.9	84,840	9,895	18,754	**75,981**	1,039,009
24	2050	532.7	319.6	213.1	86,191	9,856	19,129	**76,917**	1,115,926
25	2051	530.6	318.4	212.2	87,563	9,816	19,511	**77,868**	1,193,794
25 years		**13,924**	**8,355**	**5,570**	**1,822,886**	**257,602**	**419,349**		**1,193,794**

How each column is calculated

1. **Energy** = 512.05 kWp × 1,140.9 kWh/kWp × $(1 - 0.4\ \%)^{n-1}$
2. **Split**: 60 % used on site, 40 % exported
3. **Savings** = self-used kWh × 17.1 ct × 1.02^{n-1}
4. **Export** = exported kWh × 5.0 ct × (1 − 7.5 %)
5. **Costs** = 12,131 € × 1.02^{n-1}, plus 30,800 € for new inverters in year 13. Year-1 cost = 1.5 % × 467,345 € + 10 €/kWp × 512 kWp
6. **Net** = savings + export − costs; year 0 = −467,345 € investment

How the results follow

Payback — cumulative turns positive between year 7 and 8	**7.7 yr**
NPV = Σ $\text{Net}_n \div 1.05^n$	**442 k€**
LCOE = discounted costs ÷ discounted energy	**8.7 ct**
IRR — discount rate at which NPV = 0	**12.7 %**

payback year inverter replacement

All values from the Excel model (Economics Analysis sheet); every input can be changed there and the table updates.

What a construction-ready design would still need

01 Short string on INV01

SLD Rev. A adds 7 modules (3.3 kWp) as a separate string so all 1,085 drawn modules are used. At ~294 V hot it sits below the 500 V MPPT range: remove it, or simulate it as its own sub-array and confirm against the inverter data sheet.

02 AC breaker margin

I_b 158.8 A on a 160 A MCCB leaves almost no margin. Check the breaker's continuous rating and temperature derating, or step up to 200 A with a matching cable.

03 DC/AC optimisation

Clipping costs 3.7 % of yield. A sweep of DC/AC 1.10–1.25 in PVsyst would show the yield-versus-cost optimum.

04 Grid connection

The LV connection at the main board is an assumption. At 440 kW the DSO decides the connection point; VDE-AR-N 4110 and NA protection apply.

05 Structure & fire safety

Roof load reserve, ballast and wind uplift, plus fire-safety clearances, need a structural check before the layout is final.

06 Measured load profile

The economics (sheets 13–14) rest on an assumed 60 % self-consumption. A 15-minute load profile from the site would replace that assumption with real data.

SKILLS SHOWN

- Roof layout & row spacing
- String & inverter sizing
- Cable sizing (IEC)
- PVsyst simulation
- 3D near-shading scene
- AutoCAD · Blender
- Single-line diagrams
- LCOE · NPV · IRR

AVAILABLE ON REQUEST

A Full PVsyst simulation report (9 pages)
B Hand calculations (13 pages)
C AutoCAD layout with cable routes
D Single-line diagram, Rev. A
E Economic calculator (Excel)

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